

Heard of Laser ?

Now meet the Spacer. Essentially a laser but one that is just 44 nanometers wide. The spacer could one day replace transistors.

Faster Internet

A new protocol called SPDY is proposed that could help reduce latency. Initial experiments have shown 64 % faster load times. Read the white paper at: <http://bit.ly/digitspdypd>

Faster computing Redux

In our “Speed up your life” December 2010 issue, we introduced you to a few concepts from cutting-edge science that are poised to speed up computing like never before. In this follow-up, we take you further up the computing tunnel to bring you updates on technologies that are closer at hand and the ways in which they aim to counter the ever impending bottleneck cataclysm

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At the heart of computing lie a few key components such as microprocessors, memory and storage. Over the years, scientists have been pushing the laws of physics to make these components do much more, at a much faster rate. Let's consider microprocessors – they can perform fantastic calculations because of the sheer number of silicon transistors that are packed into them. To give you an idea, consider the first CPU on a single chip – Intel's 4004. It had approximately 2,300 transistors. From there, chip-makers shrunk manufacturing processes to crunch in more transistors on to a chip, till finally the i7 quad-cores of recent times contain 731 million transistors. But as the transistors are getting smaller, the problems faced by scientists are getting bigger.

Moore's Law, which most of us are familiar with by now, states that the number of transistors on a chip will roughly double every two years. So far



the prediction holds true like clock-work. But let's look at transistors, first, to understand what these inevitable problems of the future are. At a basic level, a transistor can be thought of as a small On / Off switch. In the On state, it lets current flow; while, in the off state, it breaks the circuit. When one transistor (that represents either a one or zero, depending on its current state) works in conjunction with a million others, the chip is able to deal with huge amounts of data at amazingly fast

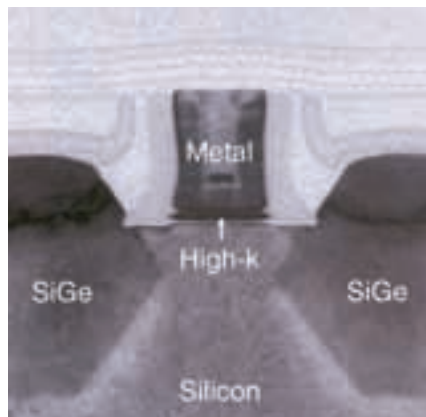


Image of Intel's transistors with Hafnium-based high-k metal gate silicon

speeds. Micro-transistors such as these are made possible because of a peculiar property of silicon (as a semiconductor) to carry current not as well as proper conductors such as metals. But silicon, too, has limitations. When you shrink transistor sizes to such unimaginable levels, there are issues with power consumption, heating, leakages, etc that creep in. In his book titled 'Fundamentals of Nanoelectronics', George Hanson makes a startling prediction that puts the problem into perspective. “With the current technologies, if the rate of increasing device density were to continue, microprocessors would soon be producing more heat per square centimeter than the surface of the sun!”, he says. There are ultimately only so many transistors, so many cores that you can pack on to a single chip. This is where scientists are turning to solutions with using materials other than silicon and efforts are even ongoing to radically improve or entirely move away from the current CMOS (complimentary metal-oxide-semiconductor) based technology on which we're now operating.